

The Electron Lattice as a Dielectric:

a periodic pinning potential for RealQM, and why it does not become a Frenkel–Kontorova model

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Status: a plan, with two measurements already made and one route already closed.

The earlier version of this document proposed measuring the two Frenkel–Kontorova coefficients and inferring the hopping barrier from them. One of the two came out cleanly; the other does not exist in this construction, for a structural reason set out below. What remains is a smaller and better-founded claim.

What is established

A geometry in which the control is exact. A rigid slide of a finite open chain is not a symmetry operation — it drags charge against the box and builds a dipole — so $E(d)$ is not periodic and no chain length repairs it. Lengthening the chain from seven cores to eleven made the asymmetry *worse*, not better. The repair is to make the translation cyclic: every electron advances one site, the rightmost leaves, a new one enters at the left, and the configuration is restored. No periodic Coulomb is needed for this, only a periodic *labelling*,

$$\text{dom} = \left\lfloor \frac{x - x_0 + a/2 - da}{a} \right\rfloor + \text{round}(d) \pmod{N},$$

where the added $\text{round}(d)$ makes the label the index of the core the domain *contains* rather than a running counter — without it the solver seeds each domain’s density at the wrong nucleus and the energies run below the physical floor.

The resulting scan, and its controls. Seven cores at the calibrated $r_c = 1.93$ and its own equilibrium spacing $a = 5.5 a_0$, the pattern slid through one full period:

d	0.000	0.125	0.250	0.375	0.500	0.625	0.750	1.000
E	−1.63252	−1.66402	−1.71667	−1.78418	−1.84151	−1.78416	−1.71666	−1.63252

$E(1) = E(0)$ to every digit stored, and all three reflection pairs agree to $1\text{--}2 \times 10^{-5}$ Ha. The noise floor is therefore about 0.5 meV against an amplitude of 5687 meV: the control passes by four orders of magnitude, where the best attempt in the companion article managed $24\times$ and six of eight failed outright.

The two numbers. The minimum is at $d = \frac{1}{2}$ — walls on cores, the bond-centred registry, independently reproducing the preference found by another route — and the maxima are at the commensurate registry:

$$\begin{aligned} \text{amplitude } V_0 &= E(0) - E(\tfrac{1}{2}) = 0.20899 \text{ Ha} = 812 \text{ meV per electron,} \\ K &= \frac{E(0.375) - 2E(0.5) + E(0.625)}{\delta u^2} = 0.2426 \text{ Ha/a}_0^2 \text{ (total)} = 0.0347 \text{ Ha/a}_0^2 \text{ per electron.} \end{aligned}$$

$K > 0$ is a restoring force: a field produces a bounded polarisation rather than a growing displacement, so there is no DC current. *This is an insulator established by a computed response rather than by a failed barrier.* The polarisability follows, $\alpha = 1/K \approx 29 \text{ a}_0^3$ per electron, between hydrogen's 4.5 and lithium's 164 — the right neighbourhood for a valence electron on a soft core, which is the only external check available.

What is closed, and why

Frenkel–Kontorova needs a second coefficient, the stiffness κ against non-uniform displacement, after which $W = \sqrt{\kappa/K}$ would give the kink width and $E_{\text{PN}} \sim \exp(-cW/a)$ the barrier without ever differencing distant configurations. That was the point of the exercise. It fails.

A sinusoidal displacement $u_m = A \sin(2\pi m/N)$, which closes periodically on the wrapped chain, gives $\Delta E < 0$ growing with A : $-0.0085, -0.0099, -0.0152, -0.0265$ Ha at $A = 0.25, 0.30, 0.35, 0.40$. Fitting and subtracting the known substrate part leaves $\kappa \approx -0.049 \text{ Ha/a}_0^2$, so W is imaginary and the model does not apply.

The cause is structural. In Frenkel–Kontorova u_m is a particle *position* and the spring acts between particles. Here the particles are slabs, and position and width are not independent: with walls at w_m , domain m has centre $\frac{1}{2}(w_{m-1} + w_m)$ and width $w_m - w_{m-1}$, so holding the widths equal forces $w_m = ma + \text{const}$ — a rigid slide. *Any non-uniform displacement necessarily changes domain sizes*, and a wider domain lets its electron spread and lower its kinetic energy. That gain — the delocalisation the companion article measures at 0.112 Ha for a vacancy — exceeds the Coulomb elastic cost.

So the negative κ is not a chain that resists stretching in reverse. It is the system preferring **unequal domains**, which also accounts for something already documented: free boundaries never settle to a uniform partition, because the uniform partition is not stable.

Honest limits on that number. The staircase noise is $\sim 0.01 \text{ Ha}$, so ΔE at the largest amplitude is only 2.7 times it, and the four ratios scatter between -0.11 and -0.17 . The *sign* is consistent across all four amplitudes; the magnitude is one significant figure at best. What is established is that κ is not positive, hence that the mapping does not close — not the value of a negative κ .

What the paper would therefore claim

Not a barrier, and not a Frenkel–Kontorova parameterisation. A **dielectric response computed from Coulomb**: a periodic pinning potential of 812 meV per electron with curvature $K = 0.0347 \text{ Ha/a}_0^2$ and polarisability 29 a_0^3 , measured on a geometry whose translation control is exact, together with the structural finding that the electron lattice is unstable to non-uniform distortion and therefore admits no Frenkel–Kontorova description.

The competitive position is unchanged and should be stated plainly: DFT supplies dielectric constants from first principles and is calibrated to do it. What is distinctive here is that the pinning

potential is a property of an explicit partition of space into unit charges, not of a smooth density — an interpretive difference, not an accuracy one.

Open, in order of value

1. **Sub-cell interface resolution.** Everything above is limited by the staircase: the partition assigns whole cells, so displacements below one cell flip no labels and K cannot be taken nearer the minimum than $u \approx 0.34 a_0$. Two configurations 0.125 and 0.250 of a spacing apart returned identical energies to five decimals — the effect caught directly. This is the one solver change that would improve every measurement in both papers.
2. **K at a second spacing and a second r_c ,** to see whether the dielectric response is stable under the parameters, which no barrier ever was.
3. **The instability characterised.** At what wavelength is the uniform chain most unstable, and does the distortion saturate? That is a Peierls-like question and the wrapped geometry can now ask it properly.